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EXPERIMENT 10

Aim: To learn Dockerfile instructions, build an image for a sample web application using DOCKERFILE.

Theory: A Dockerfile is a text file that contains a set of instructions used to automate the process of building Docker images. It defines the environment, dependencies, and commands required to run an application inside a container. Docker uses Dockerfiles to ensure consistency, portability, and reproducibility of applications across different systems.

A Dockerfile consists of several important instructions. The FROM instruction specifies the base image (e.g., Ubuntu, Node.js), while WORKDIR sets the working directory inside the container. The COPY or ADD instructions are used to transfer application files into the container. The RUN instruction executes commands during the image build process, such as installing dependencies. The CMD or ENTRYPOINT instruction defines the default command to run the application when the container starts. Additionally, EXPOSE is used to specify the port on which the application will run.

By writing a Dockerfile for a sample web application, developers can build a custom image using the docker build command and run it as a container using docker run. This approach simplifies deployment, ensures environment consistency, and allows the application to run seamlessly across development, testing, and production environments. Docker-based workflows significantly improve scalability and efficiency in modern DevOps practices.

```
C:\Users\kashishparyani>docker run redis
Unable to find image 'redis:latest' locally
latest: Pulling from library/redis
edc4b8e535e8: Pull complete
e22d95bb4ed9: Pull complete
387e0421c8da: Pull complete
4f4fb700ef54: Pull complete
dbf833dfdfed: Pull complete
0d8ecf679ede: Pull complete
5eefc97b6afb: Download complete
048ce8724475: Download complete
Digest: sha256:1f073813b641755b70b0200da64131bbeeb4ec5b633ca67772229b49820caafa
Status: Downloaded newer image for redis:latest
Starting Redis Server
1:C 20 Apr 2026 02:32:26.356 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
1:C 20 Apr 2026 02:32:26.356 * Redis version=8.6.2, bits=64, commit=00000000, modified=1, pid=1, just started
1:C 20 Apr 2026 02:32:26.356 * Configuration loaded
1:M 20 Apr 2026 02:32:26.356 * monotonic clock: POSIX clock_gettime
1:M 20 Apr 2026 02:32:26.357 * Running mode=standalone, port=6379.
1:M 20 Apr 2026 02:32:26.357 * <bf> RedisBloom version 8.6.0 (Git=unknown)
1:M 20 Apr 2026 02:32:26.357 * <bf> Registering configuration options: [
1:M 20 Apr 2026 02:32:26.357 * <bf>     { bf-error-rate       :      0.01 }
1:M 20 Apr 2026 02:32:26.357 * <bf>     { bf-initial-size     :       100 }
1:M 20 Apr 2026 02:32:26.357 * <bf>     { bf-expansion-factor :         2 }
1:M 20 Apr 2026 02:32:26.357 * <bf>     { cf-bucket-size      :         2 }
1:M 20 Apr 2026 02:32:26.357 * <bf>     { cf-initial-size     :      1024 }
1:M 20 Apr 2026 02:32:26.357 * <bf>     { cf-max-iterations   :        20 }
1:M 20 Apr 2026 02:32:26.357 * <bf>     { cf-expansion-factor :         1 }
1:M 20 Apr 2026 02:32:26.357 * <bf>     { cf-max-expansions   :        32 }
1:M 20 Apr 2026 02:32:26.357 * <bf> ]
1:M 20 Apr 2026 02:32:26.358 * Module 'bf' loaded from /usr/local/lib/redis/modules//redisbloom.so
1:M 20 Apr 2026 02:32:26.359 * <search> Redis version found by RedisSearch : 8.6.2 - oss
1:M 20 Apr 2026 02:32:26.359 * <search> RediSearch version 8.6.0 (Git=7782b97)
1:M 20 Apr 2026 02:32:26.359 * <search> Low level api version 1 initialized successfully
1:M 20 Apr 2026 02:32:26.359 * <search> gc: ON, prefix min length: 2, min word length to stem: 4, prefix max expansions: 200, query timeout (ms): 500, timeout policy: ret
urn, oom policy: return, cursor read size: 1000, cursor max idle (ms): 300000, max doctable size: 1000000, max number of search results:  1000000, default scorer: BM25STD
```

```
C:\Users\kashishparyani>docker pull redis
Using default tag: latest
latest: Pulling from library/redis
Digest: sha256:1f073813b641755b70b0200da64131bbeeb4ec5b633ca67772229b49820caafa
Status: Image is up to date for redis:latest
docker.io/library/redis:latest

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview redis
```

```
C:\Users\kashishparyani>docker stop e64661c705ec
e64661c705ec

C:\Users\kashishparyani>docker start e64661c705ec
e64661c705ec
```

```
C:\Users\kashishparyani>docker images
                                                                                                    Info →   U   In Use
IMAGE           ID             DISK USAGE   CONTENT SIZE   EXTRA
httpd:latest    bdba5c86022f        177MB         47.6MB    U
mysql:latest    c5df04bee1a4       1.27GB          283MB    U
redis:latest    1f073813b641        204MB         55.3MB    U
ubuntu:latest   c4a8d5503dfb        119MB         31.7MB    U

C:\Users\kashishparyani>docker rmi  1f073813b641
Error response from daemon: conflict: unable to delete 1f073813b641 (must be forced) - image is being used by stopped container afeca93426f5

C:\Users\kashishparyani>docker rmi 1f073813b641 -f
Untagged: redis:latest
Deleted: sha256:1f073813b641755b70b0200da64131bbeeb4ec5b633ca67772229b49820caafa
```

```
C:\Users\kashishparyani>docker run --name ubuntu --rm -i -t ubuntu bash
root@82bd269ff6ef:/# exit
exit
```

```
C:\Users\kashishparyani>docker run -d -p 8080:8080 ubuntu
74a4cf584f596220fcd63f74201e02e52e96545ae6dae50ff2f5450d7b69f610

What's next:
    Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: ports are not available: exposing port TCP 0.0.0.0:8080 -> 127.0.0.1:0: listen tcp 0.0.0.0:8080: bind: Only one usage of each socket a
ddress (protocol/network address/port) is normally permitted.

Run 'docker run --help' for more information

C:\Users\kashishparyani>
```

Conclusion: Successfully implemented Dockerfile instructions, build an image for a sample web application using DOCKERFILE.